

① Multiple Linear Regression Implementation :-

① Import Libraries - `pd, np, plt.`



② Import Dataset - `df = pd.read_csv('...')`



③ EDA -

- Null value treatment
- Remove Duplicates
- Handle categorical data.



`from sklearn.model_selection import train_test_split`

④ Split Dataset - `x = df.drop('target', axis = 1)`
`y = df['target']`



`x_train, x_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 42)`



`from sklearn.linear_model` $\Rightarrow$ Linear Regression

⑤ Training model

`regressor = LinearRegression()`
`regressor.fit(x_train, y_train)`



⑥ Result Prediction -

`y_pred = regressor.predict(x_test)`



`from sklearn.metrics` $\Rightarrow$ `r2_score`, `mean_squared_error`

⑦ Model Evaluation - Find model's goodness of fit.

`r2 = r2_score(y_test, y_pred)`

`sq_err = mean_squared_error(y_test, y_pred)`